

$$1 - P_B = \begin{bmatrix} 0.5 & 0.5 & 0.0 & 0.0 \\ 0.5 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.25 & 0.25 \end{bmatrix} \quad P_G = \begin{bmatrix} 0.0 & 0.0 & 0.5 & 0.5 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.0 & 0.0 & 0.5 & 0.5 \end{bmatrix}$$

$$V_{B,G} = G = \begin{bmatrix} 0.4 & 0.4 \\ 0.4 & 0.4 \\ 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix} \quad C_B = \begin{bmatrix} 1 & 5 & 0 & 1 \end{bmatrix}^T$$

$$C_G = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix}^T$$

LS	B	1
LC	G	2
HS	B	0
HC	B	1
States	$\Pi(n)$	Cost

$$P^\pi = \begin{bmatrix} 0.5 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.25 & 0.25 \end{bmatrix}$$

$$C^\pi = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix}^T$$

	LS	LC	HS	HC
$I - P^\pi$	0.5	-0.5	0.0	0.0
	0.0	1	-0.5	-0.5
	0.0	0.0	0.5	-0.5
	-0.25	-0.25	-0.25	0.75

$$LS \rightarrow P_{B, row 1} = \begin{bmatrix} 0.5 & 0.5 & 0.0 & 0.0 \end{bmatrix}$$

$$LC \rightarrow P_{G, row 2} = \begin{bmatrix} 0.0 & 0.0 & 0.5 & 0.5 \end{bmatrix}$$

$$HS \rightarrow P_{B, row 3} = \begin{bmatrix} 0.0 & 0.0 & 0.5 & 0.5 \end{bmatrix} \quad (V(HC) = 0)$$

$$HC \rightarrow P_{B, row 4} = \begin{bmatrix} 0.25 & 0.25 & 0.25 & 0.25 \end{bmatrix}$$

$$(I - P^\pi)V = C^\pi \rightarrow \begin{cases} V(HS) \Rightarrow 0.5V(HS) - 0.5V(HC) = 0 \Rightarrow V(HS) = 0 \\ V(LC) \Rightarrow V(LC) = 2 \\ V(LS) \Rightarrow 0.5V(LS) - 2 = 0 \Rightarrow V(LS) = 4 \end{cases}$$

$$V(LS) = 4, V(LC) = 2, V(HS) = 0, V(HC) = 0$$

$$Q(n, a) = C(n, a) + \sum_n P(n' | n, a) \cdot V(n')$$

$$\downarrow$$

$$Q(LS, B) = C_B(LS) + P_B(LS \rightarrow LS) \cdot V(LS) + P_B(LS \rightarrow LC) \cdot V(LC) = 1 + 0.5 \times 4 + 0.5 \times 2 = 4$$

$$Q(LS, G) = 2 + 0.5 \times 0 + 0.5 \times 0 = 2 \quad , Q(LS, G) < Q(LS, B) \text{ (suboptimal)}$$

$$Q(LC, G) = 2 \quad , Q(LC, B) = 5 + 0.5 \times 4 + 0.5 \times 2 = 8, Q(LS, B) < Q(LC, B) \text{ (optimal)}$$

$$Q(HS, B) = 0 \quad , Q(HS, G) = 2, Q(HS, G) > Q(HS, B) \text{ (optimal)}$$

$$Q(HC, B) = 1 + 0.25 \times 4 + 0.25 \times 2 + 0.25 \times 0 + 0.25 \times 0 = 2.5$$

$$Q(HC, G) = 2 \quad , Q(HC, G) < Q(HC, B) \text{ (suboptimal)}$$

It's not optimal policy (LC, HS are correct but fails at LS, HC)

(Note: using a discounted formula with $\gamma = 0.99$ and inserting $(I - \gamma P^\pi)^{-1} C^\pi$ shows suboptimal action only in LS. Discounting reduces the weight of future costs, putting HC's decision back in favour of battery ($J = [83.377, 84.061, 82.061, 83.719]^T$))

$$2 - b'(s) \propto \mathcal{O}(ds|s) \cdot \sum_n P(n|s, a) \cdot b(s)$$

$\mathcal{O}(ds|s) = 0.5$ for all states $\rightarrow$ state is equally likely to produce either reading; so

$$b'(s) \propto \sum_n P_G(n|s) \cdot b(s)$$

↓ computing

$$b'(LS) = \sum_n P_G(LS|s) \cdot b(s) = 0 \quad b'(LC) = 0$$

$$b'(HS) = 0.5 (b(LS) + b(LC) + b(HS) + b(HC)) = 0.5$$

$$b'(HC) = 0.5$$

$$b' \propto [0.0 \quad 0.0 \quad 0.5 \quad 0.5] \quad \text{(normalizing)} \quad b' = [0.0 \quad 0.0 \quad 0.5 \quad 0.5] = [b_1 \quad b_1 \quad b_2 \quad b_2]$$

We cannot apply the policy in the last question, because we don't know the true state, only the belief. π needs the true state to decide different actions (B or G). $b' = [0.0 \quad 0.0 \quad 0.5 \quad 0.5]$ proves that we know we're in a H state, but we don't know if it's HS or HC, and $\pi(BS) \neq \pi(HC)$, which is going to result in a suboptimal choice, as the observation doesn't tell us something new.

Exercise 3: POMDP Heuristic Policies

Setup

We are given the sequence of actions and observations: first we take action **G** and observe **S**, then we take action **B** and observe **C**. We need to compute the resulting belief state and then determine the recommended action under three heuristics: MLS, AV, and Q-MDP.

We assume a uniform initial belief: $b_0 = (0.25, 0.25, 0.25, 0.25)$ over states (LS, LC, HS, HC).

We also need the optimal MDP policy from Exercise 1. Solving the MDP with $\gamma = 0.9$, the optimal policy turns out to be $\pi^* = (\mathbf{G}, \mathbf{G}, \mathbf{B}, \mathbf{B})$, i.e., use the Grid when battery is Low and the Battery when battery is High.

Optimal Q-values from Exercise 1

The optimal value function and Q-values (with $\gamma = 0.9$) are:

State	$Q^*(s, B)$	$Q^*(s, G)$	$\pi^*(s)$	$V^*(s)$
LS	9.08	8.98	G	8.98
LC	13.08	8.98	G	8.98
HS	6.98	8.98	B	6.98
HC	8.53	8.98	B	8.53

Step 1: Belief update after action G, observation S

The POMDP belief update formula is:

$$b'(x') \propto O(x', z) \cdot \sum_x P(x' | x, a) \cdot b(x)$$

Prediction step: Compute the predicted belief after action G:

$$\hat{b} = b_0 \cdot P_{\mathbf{G}} = (0, 0, 0.5, 0.5)$$

Since every row of $P_{\mathbf{G}}$ transitions to states HS and HC with equal probability (0.5 each), the predicted belief places all mass on the High battery states.

Observation update: Incorporate observation S using $O(\cdot, S) = (0.6, 0.4, 0.6, 0.4)$:

$$b'(\text{LS}) \propto 0.6 \times 0 = 0$$

$$b'(\text{LC}) \propto 0.4 \times 0 = 0$$

$$b'(\text{HS}) \propto 0.6 \times 0.5 = 0.30$$

$$b'(\text{HC}) \propto 0.4 \times 0.5 = 0.20$$

Normalizing (dividing by the sum 0.50):

$$\mathbf{b}_1 = (0, 0, 0.6, 0.4)$$

Step 2: Belief update after action B, observation C

Prediction step: From $\mathbf{b}_1 = (0, 0, 0.6, 0.4)$, compute the predicted belief after action B:

$$\hat{b}(\text{LS}) = 0 \cdot 0.5 + 0 \cdot 0.5 + 0.6 \cdot 0.0 + 0.4 \cdot 0.25 = 0.1$$

$$\hat{b}(\text{LC}) = 0 \cdot 0.5 + 0 \cdot 0.5 + 0.6 \cdot 0.0 + 0.4 \cdot 0.25 = 0.1$$

$$\hat{b}(\text{HS}) = 0 \cdot 0.0 + 0 \cdot 0.0 + 0.6 \cdot 0.5 + 0.4 \cdot 0.25 = 0.4$$

$$\hat{b}(\text{HC}) = 0 \cdot 0.0 + 0 \cdot 0.0 + 0.6 \cdot 0.5 + 0.4 \cdot 0.25 = 0.4$$

$$\hat{b} = (0.1, 0.1, 0.4, 0.4)$$

Observation update: Incorporate observation C using $O(\cdot, C) = (0.4, 0.6, 0.4, 0.6)$:

$$b'(LS) \propto 0.4 \times 0.1 = 0.04$$

$$b'(LC) \propto 0.6 \times 0.1 = 0.06$$

$$b'(HS) \propto 0.4 \times 0.4 = 0.16$$

$$b'(HC) \propto 0.6 \times 0.4 = 0.24$$

Normalizing (dividing by the sum 0.50):

$$\mathbf{b}_2 = (0.08, 0.12, 0.32, 0.48)$$

Heuristic 1: MLS (Most Likely State)

The MLS heuristic selects the action that is optimal for the state with the highest belief probability.

From $\mathbf{b}_2 = (0.08, 0.12, 0.32, 0.48)$, the most likely state is **HC** with probability 0.48.

The optimal action for state HC is $\pi^*(HC) = B$.

Therefore, the **MLS heuristic chooses action B** (use Battery).

Heuristic 2: AV (Action Voting)

The AV heuristic has each state vote for its optimal action, weighted by belief probability. The action with the highest total vote is selected.

Votes for **B** (states where $\pi^*(s) = B$): $b(HS) + b(HC) = 0.32 + 0.48 = \mathbf{0.80}$

Votes for **G** (states where $\pi^*(s) = G$): $b(LS) + b(LC) = 0.08 + 0.12 = \mathbf{0.20}$

Since $0.80 > 0.20$, the **AV heuristic chooses action B** (use Battery).

Heuristic 3: Q-MDP

The Q-MDP heuristic selects the action that minimizes the expected Q-value under the current belief:

$$a^* = \arg \min_a \sum_x b(x) \cdot Q^*(x, a)$$

Computing the expected Q-value for each action:

$$\begin{aligned} Q(\mathbf{b}_2, \mathbf{B}) &= 0.08 \times 9.08 + 0.12 \times 13.08 + 0.32 \times 6.98 + 0.48 \times 8.53 \\ &= 0.727 + 1.570 + 2.234 + 4.095 = \mathbf{8.625} \end{aligned}$$

$$\begin{aligned} Q(\mathbf{b}_2, \mathbf{G}) &= 0.08 \times 8.98 + 0.12 \times 8.98 + 0.32 \times 8.98 + 0.48 \times 8.98 \\ &= \mathbf{8.980} \end{aligned}$$

Since $Q(\mathbf{b}_2, \mathbf{B}) = 8.625 < Q(\mathbf{b}_2, \mathbf{G}) = 8.980$ (lower cost is better), the **Q-MDP heuristic chooses action B** (use Battery).

Conclusion

All three heuristics agree and recommend action **B** (use Battery). This is consistent with the belief state $\mathbf{b}_2 = (0.08, 0.12, 0.32, 0.48)$, which places 80% of the probability mass on High battery states (HS and HC), where using the battery is the cheaper option.

Heuristic	Action
MLS	B (Battery)
AV	B (Battery)
Q-MDP	B (Battery)